

Austin Baty

Curriculum Vitæ

University of Illinois Chicago, SES Room 2258
845 West Taylor St., Chicago, IL 60607
✉ abaty@uic.edu
🌐 abaty.github.io
ORCID: 0000-0001-5310-3466

Education and Research Experience

University of Illinois Chicago

Assistant Professor 2023 -
UIC High Energy Nuclear Physics Group (CMS, ePIC Collaborations; e^+e^- Alliance)

Rice University

Research Scientist 2023
Rice Academy Postdoctoral Fellow — Advisor: Prof. Wei Li 2019-2023
Rice Relativistic Heavy Ion Group (CMS, H1 Collaborations)

Massachusetts Institute of Technology

Ph.D. in Physics 2013-2019
Research Assistant — Advisor: Prof. Gunther Roland
MIT Relativistic Heavy Ion Group (CMS Collaboration)

Texas A&M University

B.S. in Physics and Mathematics, Summa Cum Laude, University Honors 2009-2013
Research Assistant — Advisor: Prof. Peter McIntyre 2012-2013
Texas A&M Accelerator Research Lab

Research Highlights

- Led the 2025 CMS oxygen-oxygen (OO) data collection and analysis effort, resulting in the observation of jet quenching effects in light ion collisions. This was the first paper examining OO collisions, was a PRL editors' suggestion, and was recognized in *Physics Magazine*.
- Led an examination of quarkonia suppression, resulting the first observation of charged-particle multiplicity dependence of excited state production in pA collisions. Published in PRL.
- Analyzer for measurement observing QGP-like effects in single jets from pp collisions. Observes deviations from models which suggest new connections between nuclear and high-energy physics. Published as a PRL Editors' Suggestion.
- Main analyzer of a precision Z boson measurement in PbPb collisions, published in PRL. Proposes a method to remove modeling uncertainties and biases in LHC measurements.
- Main analyzer of the CMS Collaboration's 2015 charged particle nuclear modification factor paper. It has been cited over 400 times and had a figure featured in the CERN Courier.
- Author on a paper searching for collectivity in e^+e^- collisions using archived 30-year-old data from LEP. Provides a strong argument for open-data and data archival initiatives.
- Award-winning contributions to the CMS tracking detector, including leadership of the heavy ion tracking group for 7 years and a nearly 10x increase in heavy ion DAQ rate compared to 2012.

———— Awards and Honors

Breakthrough Prize in Fundamental Physics (as part of CMS)	2025
Texas A&M 12 Under 12 spotlight	2024
<i>Awarded to 12 alumni who have graduated less than 12 years ago. Given for going “...above and beyond the call of duty for students who need advocates...”</i>	
Global Young Scientist Summit (GYSS) attendee	2022
CMS Collaboration Award	2020
<i>Given “for his development work on the heavy ion track reconstruction and research activity aiming to evolve the tracker data storage scheme.”</i>	
Best talk award at 2019 USA LHC Users’ Association meeting	2019
Rice University Academy Postdoctoral Fellowship	2019
MIT School of Science Fellowship	2013-2016
Texas A&M Best Undergraduate STEM thesis	2013

———— Research Funding

Heavy ion nuclear physics research at the University of Illinois at Chicago.	2025-2028
(US Department of Energy, Office of Science; \$1,800,000; co-PI)	

Leadership Positions

LPCC Heavy Ion Working Group Convener 2023 -

This working group provides a forum for the Heavy Ion community to discuss priority topics. As a convener I organize public forums and topics of discussion.

ePIC Collaboration Council representative for UIC 2023 -

I represent UIC's interests in high-level meetings used to dictate the organization and operation of the Collaboration.

CMS Publication Committee member 2025 -

Responsible for reviewing and fine-tuning the messages and presentation of final heavy ion papers before they are sent to a journal for peer-review.

CMS Oxygen Collisions Working Group Convener 2025

Responsibilities include arranging meetings and arranging priorities for new Oxygen collision data collection. Coordinated multiple fast-track physics analyses after the data was taken.

CMS Heavy Ion Heavy Flavor and Spectra group leader 2022 - 2025

Responsibilities include arranging meetings and providing analysis advice, reviewing new analyses to maintain quality, and arranging priorities for new data collection.

Project Manager - CMS MTD ETL Nuclear Physics Consortium 2023

I coordinated a group of 5 institutions (KU, MIT, ORNL, Rice, UIC) during the construction of this \$18.4M timing detector that will be installed in CMS. I am responsible for the project being executed on-time and on-budget.

CMS Raw' Data Format Task Force Leader 2020 - 2023

The group investigated new data formats to optimize event throughput. Leadership involved managing different skill sets, and communication to ensure proper integration. Ultimately resulted in a 35% increase in number of events gathered for the LHC Run 3.

CMS Heavy Ion Tracking Contact 2016 - 2022

I coordinated 5-10 people working on tracking software. Responsibilities include mentoring students, organization, troubleshooting, and facilitating collaboration with other groups.

Others Skills and Certifications

Certified Associate in Project Management (CAPM) 2023

A globally recognized credential from the Project Management Institute indicating training in professional project management. Skills tested include project leadership, ability to control the cost, schedule, and scope of large projects, and the ability to manage various stakeholder interests.

CMS Certified Language Editor 2018 -

Indicates strong technical writing skills, and proficiency with L^AT_EX. I edit articles for the collaboration before they are submitted to a journal.

Scientific service work

Reviewer for DoE Graduate Student Research (SCGSR) Program	2024 -
Reviewer for PRC, PRD, PRL, and PLB	2022 -
Advocate for Congressional funding of physics	2020, 2021, 2025 -
<i>I have been a member of the contingent for the US LHC Users' Association and Nuclear Physics Day that advocates for NSF and DoE funding. I set up meetings with members of Congress to update them about our scientific mission and remind them of the importance of their continued support.</i>	
Conference Organizer	
<i>'53rd International Symposium on Multiparticle Dynamics,' Corfu, Greece.</i>	2025
Conference Session Chair	
<i>'22nd International Conference on Strangeness in Quark Matter,' Los Angeles, California.</i>	2026
<i>'4th Workshop on QCD Collectivity at the Smallest Scales,' Qingdao, China.</i>	2024
<i>'Opportunities of OO and pO collisions at the LHC' workshop at CERN.</i>	2021
Poster Session Judge	
<i>'8th International Conference on the Initial Stages in High-Energy Nuclear Collisions.'</i>	2025

UIC Service Work

UIC Physics Department Student Success Committee: Aug. 2025 -
UIC Physics Department Undergraduate Mentoring Committee: Aug. 2024 -
UIC Honors College Faculty Fellow: Aug. 2024 -
Physics Department Commencement Marshall: 5/5/2024
UIC Undergraduate Research Forum Judge: 4/15/2024, 4/21/2025, 4/21/2026
UIC Honors College Fall Research Symposium Judge: 11/20/2024
LAS Faculty Meeting Quorum member: Aug. 2023 -

Doctoral Thesis Defense Committee Member

I have served on the following students' committees:

*Ziyue Zhang - 5/5/2026
David Neto - 10/23/2025
Shirsendu Nanda - 10/2/2025
Gabe Dale-Gau - 6/2/2025
Jieun Yoo - 3/17/2025*

Doctoral Preliminary Exam Committee Member

I have served on the following students' committees:

*Alexei Tam - 4/30/2026
Vipul Pant - 4/30/2026
Vaibhav Lohia - 4/30/2026
Farhan Abid - 10/28/2025
Nathan Brandman-Hughes - 8/13/2025
Aaron Wang - 8/13/2025
Frank Strug - 8/13/2025
Toan Phan - 8/13/2025
Alex Thielan - 5/7/2025
Jeetendra Gupta - 5/7/2025
Aaron Wang - 5/7/2025*

Enea Prifti - 5/7/2025
Jongho Lee - 5/6/2025
Danush Shekar - 5/6/2025
Navjit Singh - 5/5/2025
Harshul Gupta - 5/5/2025
Zheng Huang - 5/5/2025
Nicholas Barnett - 5/5/2025
Abhishek Hingrajiya - 2/6/2025
Shirsendu Nanda - 5/9/2024
Ricardo Escobar - 3/29/2024
Gavin Wilks - 12/19/2023
Clayton Bennett - 12/4/2023
Ziyue Zhang - 9/22/2023

Teaching

Classes taught at UIC

PHYS 110 (Gateway to physics: Seminar for freshmen)
Spring 2026
PHYS 141 (Calculus-based introductory mechanics)
Spring 2024, Spring 2025
PHYS 392 (Physics research for undergraduates)
Spring 2026
PHYS 481 (Upper-level experimental laboratory class)
Fall 2024

Mentoring

Postdoctoral advisor

<i>Sruthy Jyothi Das:</i>	<i>2025 -</i>
<i>Prottoy Das:</i>	<i>2024 -</i>

PhD advisor

<i>Jeetendra Gupta:</i>	<i>2024 -</i>
<i>Vipul Pant:</i>	<i>2024 -</i>
<i>Zheng Huang:</i>	<i>2023 -</i>
<i>Shirsendu Nanda: (continued as a postdoc at LBNL)</i>	<i>2023 - 2025</i>
<i>Nicholas Barnett:</i>	<i>2023 -</i>

Undergraduate research advisor

<i>Leo D'Andrea: UIC undergraduate</i>	<i>2025-</i>
<i>Raven Lee: Summer intern with CMS SPRINT/PURSUE program</i>	<i>2024</i>
<i>Kyle Sabo: UIC Honors student - graduated with Honors capstone project</i>	<i>2024 - 2025</i>

While at Rice

<i>Parker Gardner: Rice graduate student</i>	<i>2020 - 2023</i>
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While at MIT

<i>Anthony Badea: MIT undergraduate student</i>	<i>2018 - 2019</i>
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Publications

As a member of the CMS Collaboration, I am a listed author on over 1000 publications. A full list can be found at [inspireHEP](#). Items here are ones I made notable contributions to. Being primary author indicates I was one of the main contributors to the analysis, while secondary author indicates I provided support in the form of corrections, cross-checks, software, etc.. Paper editing contributions usually involved significant editing/rewrites of the paper.

As a primary author

- April 2026 The CMS Collaboration. *Observation of suppressed charged-particle production in ultrarelativistic oxygen-oxygen collisions*. [PRL 136 \(2026\) 162301](#) (editors' suggestion, featured in *Physics Magazine*).
- April 2026 Das, S. J. and Baty, A.. *Data-driven method to estimate contamination from light ion beam transmutation at colliders*. [Phys. Rev. C 136 \(2026\) 162301](#).
- March 2026 The CMS Collaboration. *Observation of the jet diffusion wake using dijets in heavy ion collisions*. [Submitted to PRL](#).
- Feb. 2026 The CMS Collaboration. *System-size dependence of charged-particle suppression in ultrarelativistic nucleus-nucleus collisions*. [Submitted to PLB](#).
- Oct. 2025 The Electron-Positron Alliance. *Unbinned measurement of thrust in e^+e^- collisions at $\sqrt{s} = 91.2$ with ALEPH archived data*. [Submitted to PRL](#).
- Sept. 2025 The CMS Collaboration. *Observation of the charged-particle multiplicity dependence of $\sigma_{\Psi(2S)}/\sigma_{J/\Psi}$ in pPb collisions at 8.16 TeV*. [PRL 135 \(2025\) 092301](#).
- April 2025 The CMS Collaboration. *Overview of high-density QCD studies with the CMS experiment at the LHC*. [Phys. Rept. 1115 \(2025\) 219](#).
- Feb. 2025 The CMS Collaboration. *Pseudorapidity distributions of charged hadrons in PbPb collisions at $\sqrt{s_{NN}} = 5.36$ TeV*. [Phys. Lett. B 861 \(2025\) 139279](#).
- Nov. 2024 The CMS Collaboration. *Constraining nPDFs using dijet production in pPb collisions at 8.16 TeV with the CMS experiment*. [CMS PAS HIN-24-014](#).
- Sept. 2024 The CMS Collaboration. *Observation of enhanced long-range elliptic anisotropies inside high-multiplicity jets in pp collisions at $\sqrt{s} = 13$ TeV*. [PRL 133 \(2024\) 142301](#) (editors' suggestion).
- June 2024 The CMS Collaboration. *Extracting the speed of sound in quark-gluon plasma created in ultrarelativistic lead-lead collisions at the LHC*. [Rep. Prog. Phys. 87 \(2024\) 077801](#).
- July 2023 Baty, A., Gardner, P., and Li, W. *Novel observables for exploring QCD collective evolution and quantum entanglement within individual jets*. [Phys. Rev. C 107, 064908](#).
- Aug. 2021 The CMS Collaboration. *Constraints on the Initial State of Pb-Pb Collisions via Measurements of Z-Boson Yields and Azimuthal Anisotropy at $\sqrt{s_{NN}} = 5.02$ TeV*. [PRL 127 \(2021\) 102002](#).
- Dec. 2020 The H1 Collaboration. *Search for collectivity in e-p collisions with H1*. [Conference Note](#).
- Nov. 2019 Badea, A., et al. *Measurements of two-particle correlations in e^+e^- collisions at 91 GeV with ALEPH archived data*. [PRL 123 \(2019\) 212002](#).

- June 2019 Baty, A. *Study of Parton Energy Loss in Heavy Ion Collisions using Charged Particle Spectra Measured with CMS*. [Doctoral thesis](#).
- Aug. 2018 The CMS Collaboration. *Charged-particle nuclear modification factors in XeXe collisions at $\sqrt{s_{NN}}=5.44$ TeV*. [JHEP 10 \(2018\) 138](#).
- April 2017 The CMS Collaboration. *Charged-particle nuclear modification factors in PbPb and pPb collisions at $\sqrt{s_{NN}}=5.02$ TeV*. [JHEP 1704 \(2017\) 039](#).
- Jan. 2016 The CMS Collaboration. *Measurement of transverse momentum relative to dijet systems in PbPb and pp collisions at $\sqrt{s_{NN}}=2.76$ TeV*. [JHEP 01 \(2016\) 006](#).
- June 2015 The CMS Collaboration. *Jet Fragmentation Function in pPb Collisions at $\sqrt{s_{NN}}=5.02$ TeV and pp Collisions at $\sqrt{s}=2.76$ and 7 TeV*. [CMS PAS HIN-15-004](#).
- Sept. 2013 Sooby E. et al. *Candidate molten salt investigation for an accelerator driven subcritical core*. [Journal of Nuclear Materials \(2013\) 440\(1\), 298–303](#).
- June 2013 Baty, A. *Molecular Dynamics Simulation of the Transport Properties of Molten Transuranic Chloride Salts*. [Undergraduate research thesis](#).

As a secondary author

- Feb. 2026 The CMS Collaboration. *Probing early parton emissions in heavy ion collisions using the Lund jet plane*. [Submitted to PRL](#).
- Oct. 2025 The CMS Collaboration. *Observation of long-range collective flow in OO and NeNe collisions and implications for nuclear structure studies*. [Submitted to PRL](#).
- Sept. 2025 The CMS Collaboration. *First measurement of pseudorapidity distributions of charged hadrons in oxygen-oxygen collisions at $\sqrt{s_{NN}}=5.36$ TeV with CMS*. [CMS PAS HIN-25-010](#).
- July 2025 The CMS Collaboration. *Search for jet quenching with dijets from high-multiplicity pPb collisions at $\sqrt{s_{NN}}=8.16$ TeV*. [JHEP 07 \(2025\) 118](#).
- Sept. 2024 Chen, Y.-C., et al. *Long-range near-side correlation in e^+e^- collisions at 183–209 GeV with ALEPH archived data*. [PLB 856 \(2024\) 138957](#).
- March 2023 Arslandok M., et al. *Hot QCD White Paper*. [arXiv:2303.17254](#).
- March 2023 Achenbach P., et al. *The Present and Future of QCD*. [As submitted to the 2023 NSAC Long Range Planning committee](#).
- June 2022 Chen, Y., et al. *Jet energy spectrum and substructure in e^+e^- collisions at 91.2 GeV with ALEPH Archived Data*. [JHEP 06 \(2022\) 008](#).
- June 2021 The CMS Collaboration. *First measurement of large area jet transverse momentum spectra in heavy-ion collisions*. [JHEP 05 \(2021\) 284](#).
- May 2021 The CMS Collaboration. *Measurement of b jet shapes in proton-proton collisions at $\sqrt{s}=5.02$ TeV*. [JHEP 05 \(2021\) 054](#).
- April 2019 The CMS Collaboration. *Jet shapes of isolated photon-tagged jets in PbPb and pp collisions at $\sqrt{s_{NN}}=5.02$ TeV*. [PRL 122 \(2019\) 152001](#).
- Dec. 2018 The CMS Collaboration. *Observation of medium-induced modifications of jet fragmentation in Pb-Pb collisions at $\sqrt{s_{NN}}=5.02$ TeV using isolated photon-tagged jets*. [PRL 121 \(2018\) 242301](#).
- April 2018 The CMS Collaboration. *Pseudorapidity distributions of charged hadrons in proton-lead collisions at $\sqrt{s_{NN}}=5.02$ and 8.16 TeV*. [JHEP 01 \(2018\) 045](#).

March 2016 The CMS Collaboration. *Correlations between jets and charged particles in PbPb and pp collisions at $\sqrt{s_{NN}} = 2.76$ TeV.* [JHEP 02 \(2016\) 156](#)

As the collaboration's paper editor

- Sept. 2025 The CMS Collaboration. *Probing gluon fluctuations in nuclei with the first energy-dependent measurement of incoherent J/Ψ photoproduction in ultraperipheral PbPb collisions.* [PRL 135 \(2025\) 112301](#) (editors' suggestion).
- Sept. 2025 The CMS Collaboration. *Multiplicity dependence of charm baryon and charm meson production in pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV.* [PLB 868 \(2025\) 139672](#).
- June 2025 The CMS Collaboration. *Search for medium effects using jet axis decorrelation in inclusive jets from PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV.* [JHEP 06 \(2025\) 120](#).
- Oct. 2024 The CMS Collaboration. *Correlations between azimuthal anisotropy and mean transverse momentum in pp, pPb, and peripheral PbPb collisions.* [Submitted to PRL](#).
- Dec. 2023 The CMS Collaboration. *Probing small Bjorken- x nuclear gluonic structure via coherent J/Ψ photoproduction in ultraperipheral PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV.* [PRL 131, \(2023\) 262301](#).
- Oct. 2023 The CMS Collaboration. *Measurements of the azimuthal anisotropy of charmonia in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV.* [JHEP 10 \(2023\) 115](#).
- July 2022 The CMS Collaboration. *Probing charm quark dynamics via multiparticle azimuthal correlations in 5.02 TeV PbPb collisions.* [PRL 129, \(2022\) 022001](#).
- Sept. 2021 The CMS Collaboration. *Observation of forward neutron multiplicity dependence of dimuon acoplanarity in ultraperipheral PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV.* [PRL 127, \(2021\) 122001](#).
- July 2020 The CMS Collaboration. *Measurement of quark- and gluon-like jet fractions using jet charge in PbPb and pp collisions at 5.02 TeV.* [JHEP 07 \(2020\) 115](#).

Seminars and Colloquia

- May 2026 *Exploration of Light Ion Collisions at the Large Hadron Collider*. May 14, 2026. Colloquium Lehigh University Physics Department Colloquium, Bethlehem, Pennsylvania.
- Jan. 2026 *Hot Stuff: Emergence in Nuclear Physics*. January 22, 2026. UIC Physics Club Seminar General Meeting, Chicago, Illinois.
- Feb. 2025 *Studying emergence with the strong force*. Feb. 5, 2025. UIC Faculty Colloquium, Colloquium Chicago, Illinois.
- Jan. 2025 *Exploring the complexity of trillion degree matter*. Jan. 28, 2025. Department of Colloquium Physics and Astrophysics Colloquium, DePaul University, Chicago, Illinois.
- Dec. 2024 *Quarkonia suppression in small systems at the LHC and prospects for hadronization studies at the EIC*. December 2, 2024. Nuclear Physics Seminar, University of Illinois Urbana-Champaign, Urbana, Illinois.
- Nov. 2024 *Emergence in QCD matter and the future of QCD studies at the EIC*. November Seminar 4, 2024. Experimental Particle and Astro-Particle Physics Seminar, University of Zurich, Switzerland.
- July 2024 *High-energy nuclear physics and heavy ion collisions with CMS*. July 3, 2024. CMS Seminar PURSUE/SPRINT Program Seminar, Fermilab, Chicago, IL.
- May 2024 *Speed of sound in quark-gluon plasma*. May 16, 2024. Collider cross talk seminar, Seminar CERN Department of Theoretical Physics, Geneva, Switzerland.
- May 2024 *Emergence in trillion-degree matter and other high-density systems*. May 6, 2024. Colloquium Enrico Fermi Institute Colloquium, University of Chicago, Chicago, Illinois.
- April 2024 *UIC at the LHC and EIC: The Present and Future of Nuclear Physics*. April 10, Colloquium 2024. UIC Faculty Colloquium, Chicago, Illinois.
- July 2023 *Observation of enhanced long-range elliptic anisotropies inside high-multiplicity jets in pp collisions at the LHC*. July 18, 2023. LHC Seminar, LHC Physics Center at Seminar CERN (LPCC), Geneva, Switzerland.
- April 2023 *Jets and trillion-degree matter: studying the emergence of QCD phenomena*. April 5, Colloquium 2023. University of Illinois Chicago Physics Colloquium, Chicago, Illinois.
- Jan. 2023 *Jets and trillion-degree matter: studying QCD at multiple scales*. Jan. 26, 2023. Yale Seminar University Nuclear, Particle, and Astrophysics seminar, New Haven, Connecticut.
- Feb. 2022 *Trillion Degree Matter: Probing the emergence of the quark-gluon plasma*. Feb. 3, Colloquium 2022. Vanderbilt University Department of Physics and Astronomy Colloquium, Nashville, Tennessee.
- Feb. 2022 *Trillion Degree Matter: Probing the emergence of the quark-gluon plasma*. Feb. 1, Seminar 2022. Rice University Nuclear and Particle Physics Seminar, Houston, Texas.
- Oct. 2019 *Searching for collectivity in small systems using e^+e^- data from ALEPH*, October 1, Seminar 2019. LBL Heavy Ion Tea Seminar, Berkeley National Lab, Berkeley, California.
- Feb. 2019 *Exploring charged particle production in the quark-gluon plasma: from e^+e^- to PbPb*, Seminar Feb. 7, 2019. CU Boulder, Boulder, Colorado.

Conference Presentations

Talks at larger conferences and high profile talks are indicated by bold font.

- March 2026 ***Collectivity in the smallest systems.*** March 31, 2026. Kavli Institute for
Plenary Theoretical Physics Conference: Probing our Understanding of Quark-
Gluon matter, UCSB, Santa Barbara, California.
- March 2026 *Probing Collectivity and Hadronization in OO and NeNe Collisions at the LHC.*
Parallel March 24, 2026. 22nd International Conference on Strangeness in Quark Matter,
UCLA, Los Angeles, California.
- Oct. 2025 *Nuclear physics in the next decade: Early career perspective.* Oct. 19, 2025.
Workshop RHIC/AGS Users' Executive Committee Open Forum, Chicago, Illinois.
- Sept. 2025 *Review of Ultraperipheral Collision results from ATLAS and CMS.* Sept. 25, 2025.
Plenary 53rd International Symposium on Multiparticle Dynamics, Corfu, Greece.
- Sept. 2025 ***Characterizing initial and final state effects with light ion collisions in***
Plenary ***CMS.*** Sept. 11, 2025. 8th International Conference on the Initial Stages
in High-Energy Nuclear Collisions, Taipei, Taiwan.
- Aug. 2025 ***Overview of recent heavy ion results at CMS.*** Aug. 27, 2025. 32nd
Parallel International Symposium on Lepton Photon Interactions at High Energies,
Madison, Wisconsin.
- June 2025 *Early Analysis of Oxygen/Neon Runs.* June 12, 2025. CMS Week, CERN, Geneva,
Workshop Switzerland.
- May 2025 *QCD collectivity in high-multiplicity jets from pp collisions.* May 21, 2025. 2025
Workshop RHIC/AGS Annual Users' Meeting, Brookhaven National Lab, New York.
- April 2025 ***New constraints on nPDFs using dijets in pPb collisions at 8.16 TeV***
Parallel ***with CMS.*** April 8, 2025. Quark Matter 2025, Frankfurt, Germany.
- April 2025 ***Recent highly-differential measurements examining jet quenching.*** April
Workshop 3, 2025. High Energy Probes of the Initial Stages, CERN Theory Institute,
Geneva, Switzerland.
- Sept. 2024 ***Observation of the multiplicity dependence of $\sigma_{\psi(2S)}/\sigma_{J/\psi}$ in pPb collisions***
Parallel ***and implications for comover effects.*** September 25, 2024. Hard Probes
2024, Nagasaki, Japan.
- Aug. 2024 ***Final state interactions in heavy ion collisions with CMS and at the EIC.***
Workshop August 21, 2024. Heavy Ion Physics in the EIC Era Workshop, Institute
for Nuclear Theory, Seattle, Wa..
- June 2024 ***Collectivity in jets: Experiment.*** June 25, 2024. 4th International Work-
Workshop shop on QCD Collectivity at the Smallest Scales, Qingdao, China.
- June 2024 *Measurement of the multiplicity dependence of charm hadron production in pPb*
Parallel *collisions with CMS.* June 4, 2024. 21st International Conference on Strangeness in
Quark Matter, Strasbourg, France.
- June 2024 *Long range correlations in high-multiplicity e^+e^- collisions using archived ALEPH*
Parallel *data at 91-209 GeV.* June 4, 2024. 21st International Conference on Strangeness in
Quark Matter, Strasbourg, France.
- Nov. 2023 *Extracting the speed of sound in the strongly interacting matter in ultrarelativistic*
Workshop *nuclear collisions.* Nov. 21, 2023. 14th International Workshop on Multiple Parton
Interactions at the LHC, Manchester, UK.

- Sept. 2023 *Overview of Recent CMS Results.* Sept. 4, 2023. Quark Matter 2023, Plenary Houston, Texas.
- June 2023 *Status of DOE-NP ETL Project.* June 1, 2023. US CMS Annual Collaboration Parallel Meeting, Carnegie Mellon University, Pittsburgh, Pennsylvania.
- Mar. 2023 *Overview of Recent Experimental Electroweak Probes Results.* March 27, Plenary 2023. Hard Probes 2023, Aschaffenburg, Germany.
- Oct. 2022 *Constraints on the initial state of heavy ion collisions at the LHC using electroweak bosons.* Oct. 14, 2022. Fall Meeting of the Texas Section of the APS, Houston, Texas. Parallel
- June 2022 *Overview of recent heavy ion results.* June. 29, 2022. CMS Week, CERN, Plenary Switzerland.
- April 2022 *Electroweak probes as tools for understanding initial state effects in heavy ions with CMS.* April. 7, 2022. Quark Matter 2022, Krakow, Poland. Parallel
- Aug. 2021 *Measurement of charged particle multiplicity distributions in DIS at HERA and its implication to entanglement entropy of partons.* Aug. 5, 2021. 14th Quark Confinement and the Hadron Spectrum Conference. Parallel
- Jan. 2021 *Measurement of initial stages via color neutral probes in pPb and PbPb.* Parallel Jan. 14, 2021. Initial Stages 2021.
- Oct. 2020 *New constraints of initial states in PbPb collisions with Z boson yields and azimuthal anisotropy at 5.02 TeV.* Oct. 29, 2020. APS Division of Nuclear Physics fall meeting (DNP 2020). Parallel
- Oct. 2020 *Future opportunities for exploring collectivity using extra small systems.* Oct. 28, Parallel 2020. Snowmass EF07 meeting: High Density QCD in Small Collision Systems
- June 2020 *Probing initial and final state effects with Z bosons in PbPb and Drell-Yan in pPb with the CMS detector,* June 3rd, 2020. Hard Probes 2020. Parallel
- Nov. 2019 *Electroweak Boson production in Heavy-Ion Collisions from CMS,* November 11th, Workshop 2019. A Workshop on Heavy Flavor and Dilepton Production in Relativistic Heavy-Ion Collisions (HeFe2019), University of Science and Technology of China, Hefei, China.
- Nov. 2019 *Collectivity of Heavy Flavor Hadrons in pp and pPb Collisions with the CMS Detector,* November 5th, 2019. Quark Matter 2019, Wuhan, China. Parallel
- Oct. 2019 *Using charged particles to study jet quenching in heavy ion collisions with CMS,* Parallel October 17, 2019. US LHC Users Association Annual Meeting 2019, Rice University, Houston, Texas.
- June 2019 *Searches for the ridge in DIS and e^+e^- ,* June. 27, 2019. Initial Stages Plenary 2019, Columbia University, New York City, New York.
- June 2019 *Studies of archived e^+e^- and HERA data,* June. 4, 2019. 2019 RHIC & AGS Annual Parallel Users' Meeting, Brookhaven National Lab, Brookhaven, New York.
- Dec. 2018 *Measurements of two-particle correlations in e^+e^- collisions at 91 GeV with ALEPH archived data,* Dec. 11, 2018. 10th International Workshop on Multiple Parton Workshop Interactions at the LHC, Perugia, Italy.
- Oct. 2018 *Charged Particle Nuclear Modification Factors in pPb, PbPb, and XeXe Collisions with CMS,* Oct. 2, 2018. Hard Probes 2018, Aix-Les-Bains, Parallel France.
- May 2018 *Charged Particle Nuclear Modification Factors in pPb, PbPb, and XeXe Collisions with CMS,* May 15, 2018. Quark Matter 2018, Venice, Italy. Parallel

- Sept. 2017 *Boson+Jet Correlations and Boson-Tagged Jet Substructure in pp and PbPb collisions at 5.02 TeV with CMS*, Sept 20, 2017. Initial Stages 2017, Krakow, Poland
- Sept. 2016 *Charged particle nuclear modification factor in PbPb at 5.02 TeV with CMS*, Sept 25, 2016. Hard Probes 2016, Wuhan, China
- July 2016 *Jet and Hadron R_{AA}* , July 27, 2016. 4th Heavy-Ion Jet Workshop, Paris, France Workshop
- May 2016 *Charged-particle production in PbPb and pp collisions at 5 TeV with CMS*, May 24, 2016. Initial Stages 2016, Lisbon, Portugal
- Sept. 2015 *Fragmentation patterns of jets in pPb collisions in CMS*, September 28, 2015. Quark Matter 2015, Kobe, Japan

Outreach and Relevant Public Relations Articles

- April 2026 *First signs of quark–gluon plasma in oxygen–oxygen collisions.* [Physics Magazine synopsis: online magazine by the American Physics Society, April. 22, 2026.](#)
- Nov. 2025 *UIC scientists recreate the universe’s first moments by slamming together oxygen ions.* [UIC today news release, Nov. 17, 2025.](#)
- Sept. 2025 *Collective motion in tiny systems: oxygen and neon collisions as a window into nuclear structure.* [CMS News Release, Sept. 18, 2025.](#)
- Sept. 2025 *LHC’s first neon-neon collisions – CMS spots an interesting pattern.* [CMS News Release, Sept. 16, 2025.](#)
- Sept. 2025 *LHC’s first oxygen collisions – CMS spots signs of small-scale quark-gluon plasma.* [CMS News Release, Sept. 11, 2025.](#)
- May. 2025 UIC scientists among winners of prestigious Breakthrough Prize in Fundamental Physics. [UIC Today, May. 22, 2025.](#)
- Feb. 2024 *Hearing the sound of quark-gluon plasma.* [CERN homepage News Release, February 16, 2024.](#)
- Jan. 2024 *A Big Bang from a Quantum Quark?* [CMS News Release, January 22, 2024.](#)
- Jan. 2024 *QGP production studied at record energies.* [CERN Courier Jan/Feb Issue, \(2024\) p16.](#)

Other work

Collaborating on Virtual Reality Research

2025 -

I have worked with a graduate student in the UIC Learning Sciences Research Institute (LSRI) who is researching how virtual reality (VR) environments can enhance physics education. I have lent my physics-specific expertise gained from teaching 100-level classes to help design an instructive and pedagogical VR space.